

LOW POWER FUSION PROCESSOR: DUAL CORTEX-A32, DUAL CORTEX-M55, ETHOS-U85 & U55 NPUs, SECURITY, MIPI-DSI/CSI, ISP, 2D GPU, DUAL HEXSPI, ANALOG INTERFACES, 9.75MB SRAM, 5.5MB MRAM

Features

High-Performance Quad-Core Fusion Processor

- Dual Arm® Cortex®-A32 Cores with Arm Neon™ SIMD Extension up to 800 MHz, 512KB Shared L2, 32KB L1 Instruction and Data Caches, MMU, Armv8-A ISA with Arm TrustZone®
- High-Performance (HP) Arm® Cortex®-M55 Core, up to 400 MHz, with Helium™ Vector Processing Extension, Double-Precision FPU, 1.25MB SRAM 0-wait State Tightly-Coupled Memory (TCM), 32KB Instruction and Data Caches, Armv8.1-M ISA with Arm TrustZone®, and 4.37 CoreMark®/MHz Performance Benchmark.
- High-Efficiency (HE) Arm® Cortex®-M55 Core, up to 160 MHz, with Helium™ Vector Processing Extension, Double-Precision FPU, 512KB of SRAM 0-wait State Tightly Coupled Memory, 32KB Instruction and Data Caches, Armv8.1-M ISA with Arm TrustZone®, and 4.37 CoreMark®/MHz Performance Benchmark.
- High-Performance 400-MHz 64-bit and 128-bit AXI Bus Fabric Common Across System

Efficient NPUs for Generative AI/ML Acceleration

- 1× Arm Ethos™-U85 Neural Processing Unit (NPU) for Generative AI (NPU-HG), 256 MAC/cycle up to 400 MHz and 204 GOPS, Supporting Recurrent Neural Networks (RNN), Convolutional Neural Networks (CNN), and Transformer Networks, Fed by 128-bit wide High-Bandwidth Local Memory at over 12 GB/s
- 1× Arm Ethos-U55 NPU (NPU-HP), 256 MAC/cycle up to 400 MHz and 204 GOPS, Supporting RNN and CNN Networks
- 1× Arm Ethos-U55 NPU (NPU-HE), 128 MAC/cycle up to 160 MHz and 46 GOPS, Supporting RNN and CNN Networks
- On-the-Fly Weight Decompression with Dedicated DMA Controllers

Extreme-Low Power Technology

- Autonomous Intelligent Power Management (aiPM™)

- FD-SOI Low Leakage Process
- 1.3 µA Consumed in STOP Mode with LPRTC, LPTIMER, LPCMP, BOR, 4KB Utility SRAM, Wake Pins
- As Low as 27 µA/MHz Dynamic Consumption for High-Efficiency Cortex-M55
- Multiple Power Domains, Dynamic Power Gating, Voltage and Clock Scaling, DC-DC Converter

On-Chip Application Memory

- High Endurance MRAM Non-Volatile Memory
 - 5.5MB, Minimum 100,000 Write Cycles
- SRAM
 - 9.75MB total SRAM: 8MB Bulk + 1.25MB M55-HP TCM + 0.5MB M55-HE TCM
 - Optional Data Retention of Bulk SRAM and M55-HE TCM SRAM Contents
 - 4KB Always-On Utility SRAM

External Memory Interfaces

- 2× 16-bit Hex SPI, each up to 400 MB/s SDR, 800 MB/s DDR, with Inline AES Decryption and Support for XIP and HyperBus
- 1× SD® v4.2, eMMC™ v5.1 Channel with DMA

Secure Enclave

- Hardware-based Root-of-Trust (RoT) with Unique Device ID
- Secure Key Generation and Storage, Secure Certificate Storage, Secure Boot
- In-Factory Provisioned Private Keys
- Crypto Accelerators—AES (up to AES-256), ECC (up to 384 bits), SHA (up to SHA-256), RSA (up to RSA-3072), and NIST compliant TRNG
- Secure Debugging with Certificate Authentication

Timing Control and Measurement

- 12× Universal High-Resolution 32-bit Timers Capable of Motor and LED Lighting Control
- 3× Low-Power Universal High-Resolution 32-bit Timers
- 4× Watchdog Timers
- 4× Low-Power 32-bit Timers

- 1× Real-Time Counter
- 4× Quadrature Encoder Counters

Serial Communication Interfaces

- 1× 10/100 Ethernet with DMA
- 1× USB 2.0 HS/FS Host/Device with DMA
- 1× SDIO v4.1 Channel with DMA
- 1× CAN FD Channel up to 10 Mbps
- 1× MIPI® I3C® Channels
- 1× Low-Power I3C Channel
- 4× I2C Channels up to 3.4 Mbps Throughput
- 2× Low-Power I2C Channels
- 8× UART Channels up to 5 Mbps (4× with RS-485 Driver Control)
- 1× Low-Power UART Channel
- 4× SPI Channels up to 50 Mbps Throughput
- 1× Low-Power SPI Channel

Analog Interface Capabilities

- 3× 12-bit SAR ADC (18 Single-Ended Inputs)
- 1× 24-bit Sigma-Delta ADC (4 Differential Inputs)
 - Programmable Gain Instrumentation Amplifier (1× to 128×)
- 2× 12-bit DACs (2 channels)
- 4× High-Speed Analog Comparators with 2.5-ns Response (16 Inputs)
- 1× Low-Power Analog Comparator (4 Inputs)
- Internal Temperature Sensor
- Internal Precision Reference Voltage

Camera Interfaces

- 1× 2-Lane MIPI D-PHY CSI-2®, 2.5 Gbps each lane
- Option for 2× MIPI D-PHY CSI-2 Interface by using the MIPI D-PHY DSI in Slave Mode. Allows Interleaving Two Image Streams from Two Separate Image Sensors
- 1× Camera Parallel Interface (CPI), up to 16 bits
- 1× Low-Power CPI, up to 8 bits

ISP Acceleration Unit

- Clocked up to 400 MHz, 64-bit Bus Connection
- Maximum Input/Output Resolution: 2MP (1920 × 1080)
- Configurable Frame Rate Control: 1 to 60 FPS
- Input Monochrome Formats: 8-bit, 10-bit
- Conversions: Demosaic Bayer to RGB, YUV422 to YUV444 to RGB
- Output Formats: 8-bit Monochrome, RAW 8/10/12-bit with Raster Scanning, RGB888 Planar, YUV, Y-only

- Auto White Balance and Auto Exposure
- Gamma Correction
- Crop, Scale, Binning, Black Level, Gain

JPEG Encoder

- Works in Tandem with ISP to Store Compressed Images and Motion JPEG (MJPEG) Video Clips
- Maximum Performance: 2MP at 200 FPS
- Input Formats: YUV400, YUV420, YUV422, RGB
- Supports JPEG ISO/IEC 10918-1, ITU-T T.81
- Supports MJPEG T.81 Format
- Encoding Color Space: YUV 4:0:0, 4:2:0, 4:2:2
- Cropping, Rotation, Padding

Display Interfaces

- Graphics LCD Controller
- 1× Display Parallel Interface (DPI), 24-bit RGB
- 1× 2-Lane MIPI D-PHY DSI®, 2.5 Gbps each lane

Graphics

- D/AVE 2D Graphics Processing Unit

Audio Interfaces

- 4× I2S Synchronous Stereo Audio Interfaces
- 1× Low-Power I2S Stereo Audio Interface
- 4× 2-channel Pulse Density Modulation (PDM) Microphone Inputs (8 Mono Microphones)
- 4× 2-channel Low-Power Pulse Density Modulation (LPPDM) Microphone Inputs (8 Mono Microphones)

General Input/Output

- Up to 136× 1.8-V GPIOs (Shared with Peripherals)
- Up to 8× Selectable 1.8-V to 3.3-V GPIOs (Shared with Peripherals)

Clock Generation

- LFRC - Internal Low-Frequency RC Oscillator (32.7 kHz, ±4%)
- HFRC - Internal High-Frequency RC Oscillator (Up to 76.8 MHz, ±2%)
- LFXO - External Low-Power Crystal Oscillator or Quartz Crystal (32.768 kHz)
- HFXO - External High-Frequency Crystal Oscillator or Quartz Crystal (24 MHz to 38.4 MHz)
- One Fractional-N PLL to Generate System Clock, Additional Fractional-N PLL to Generate Audio Clock

System

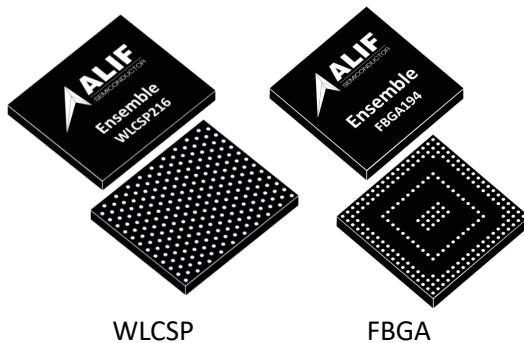
- Global Event Mapping to Configurable Triggers
- 3× 32-Channel General DMA Controllers
- CRC Calculation Accelerator with Programmable Polynomials
- Programmable Low Supply Voltage Detect Warning (Brown-Out Detect)
- Power-On Reset and Brown Out Reset
- Real-Time Clock
- JTAG/SWD Debug Interface

Operating Parameters

- 1.75 V to 4.8 V Primary Supply Range
- 1.08 V to 1.98 V I/O Supply Range (1.8 V I/O)
- 3.0 V to 4.0 V I/O Supply Range (3.3 V Flex I/O)
- -40 °C to 75 °C Ambient Temperature Range
- -40 °C to 105 °C Junction Temperature Range

Packages

- WLCSP216, 0.4 mm Pitch
- FBGA194, 0.5 mm Pitch



WLCSP

FBGA

Device Overview

The families of fusion processors and microcontrollers (MCUs) from Alif Semiconductor create a scalable and compatible continuum of highly integrated embedded processor devices for use in low-end to high-end intelligent IoT end-point applications. Architected for power efficiency and long battery life, these devices deliver high computation and ML/AI capability, multi-layered security, computer vision, and highly interactive human-machine interface.

Individual device selections within a family of devices scale up starting with single-core MCUs, dual-core MCUs, triple-core MCU/MPU fusion processors, and quad-core MCU/MPU fusion processors to match specific applications. Across all devices are common peripherals, common power management schemes, and a common interconnection fabric making it easy to re-use software and hardware over many varied projects.

This document covers the second-generation quad-core embedded fusion processors from the Ensemble™ family, designated as the E8 series.

Power efficiency—In addition to the use of FD-SOI silicon process technology, *aiPM* provides fine granular control over the processing, memory, and peripheral resources of the device, resulting in extreme power conservation that consumes only what is needed, when its needed, at any given instant based on use case.

Computation and ML/AI—Quad-core design with two Arm Cortex-A32 application cores, and two Arm Cortex-M55 Real-Time cores, with one of the Real-Time cores optimized for extreme low power operation. Machine Learning and AI tasks are accelerated by triple Arm Ethos neural processing units, generating as much as 454 GOPS in total. One NPU is the Ethos-U85 that enables RNN, CNN, and TNN transformer-based ML models for generative AI. The other two NPUs are Ethos-U55 for RNN and CNN ML models. E8 devices can operate with an RTOS, an OS such as Linux, or both, while using only the on-chip memory resources. Additional RAM and Flash memory may be accessed externally through two high-speed 16-bit Hex SPI interfaces supporting XIP mode and HyperBus protocol. An in-line AES decryption module in each Hex SPI interface ensures the confidentiality of the data and/or code stored in the external memory devices.

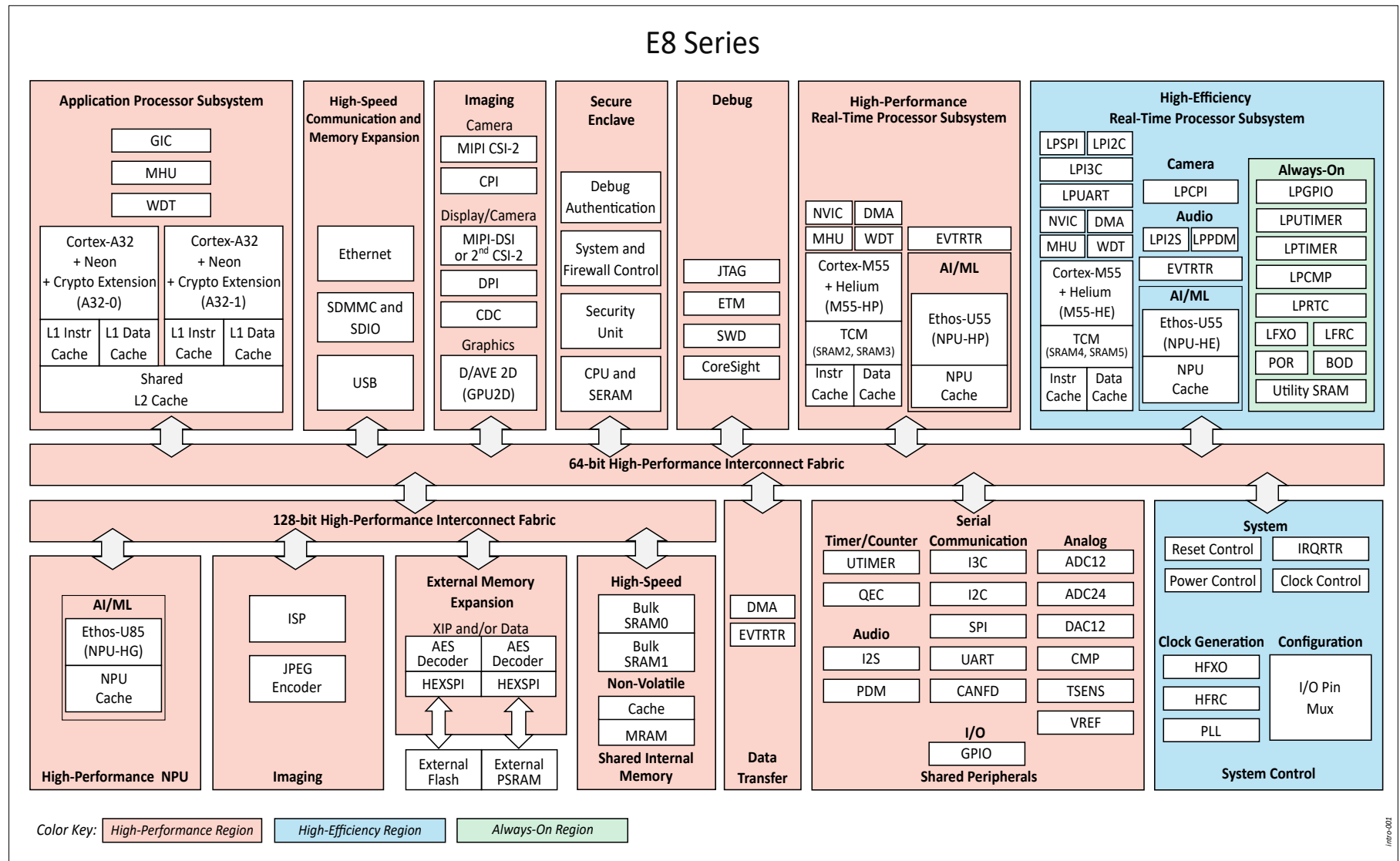
Connectivity—E8 devices support many wired interfaces including Ethernet, USB, SDIO, CANFD, I3C, I2C, and more.

Security—An isolated Secure Enclave manages the entire life cycle of the end application from manufacture, to deployment, to secure Firmware Over-The-Air updates, and to retirement. A unique Root-of-Trust existing in the enclave enables a trusted on-chip platform for key generation, secure storage, secure boot, cryptographic acceleration, and more. The Secure Enclave also enables certificate-based secure debugging.

Computer Vision—Integrated camera interfaces, including one or two MIPI CSI-2 with hardware Image Signal Processing (ISP) and JPEG compression, enable easy connection of low-power image sensors for AI image classification using the Arm Ethos NPUs. Biometric ID, face detection, object classification, barcode reading, and other vision applications can execute while consuming very little energy.

Human Machine Interface—Integrated color display interfaces, including MIPI DSI, can drive a wide range of display panels with vivid graphics supported by the on-chip 2D graphics processing unit for attractive user interfaces. Digital audio inputs and outputs are supported by PDM and I2S interfaces. The Arm Ethos NPUs can apply ML inference to audio input streams for key word spotting, contextual-aware speech recognition, failure prediction, and more.

Device Block Diagram



Ordering Information

TableOrderable Part Numbers

Part Number ⁽¹⁾	MRAM	SRAM	GPIO	Package	Operating Temperature
AE822FA0E5597BS0	5.5MB	9.75MB	144	WLCSP216	-40C to 70C
AE822FA0E5597LS0	5.5MB	9.75MB	128	FBGA194	-40C to 70C

1. Ordering designation for shipment packaging:

- Add the following suffix to base part number
 - -T, for Tape and Reel
 - -Y, for Tray
- Example: AE822FA0E5597BS0-T is Tape and Reel

Part Numbers SRAM Breakdown

Part Number	Total SRAM (KB)	SRAM0 (KB)	SRAM1 (KB)	SRAM2 (M55-HP ITCM) (KB)	SRAM3 (M55-HP DTCM) (KB)	SRAM4 (M55-HE ITCM) (KB)	SRAM5 (M55-HE DTCM) (KB)
AE822FA0E5597BS0	9984	4096	4096	256	1024	256	256
AE822FA0E5597LS0	9984	4096	4096	256	1024	256	256